Unit-1

Introduction to artificial intelligence

What is artificial intelligence?

Artificial intelligence is a specialty within computer science that is concerned with creating systems that can replicate human intelligence and problem-solving abilities. They do this by taking in a myriad of data, processing it, and learning from their past in order to streamline and improve in the future. A normal computer program would need human interference in order to fix bugs and improve processes.

The history of artificial intelligence:

The idea of "artificial intelligence" goes back thousands of years, to ancient philosophers considering questions of life and death. In ancient times, inventors made things called "automatons" which were mechanical and moved independently of human intervention. The word "automaton" comes from ancient Greek, and means "acting of one's own will." One of the earliest records of an automaton comes from 400 BCE and refers to a mechanical pigeon created by a friend of the philosopher Plato. Many years later, one of the most famous automatons was created by Leonardo da Vinci around the year 1495.

So while the idea of a machine being able to function on its own is ancient, for the purposes of this article, we're going to focus on the 20th century, when engineers and scientists began to make strides toward our modern-day AI.

Groundwork for AI:

1900-1950In the early 1900s, there was a lot of media created that centered around the idea of artificial humans. So much so that scientists of all sorts started asking the question: is it possible to create an artificial brain? Some creators even made some versions of what we now call "robots" (and the word was coined in a Czech play in 1921) though most of them were relatively simple. These were steam-powered for the most part, and some could make facial expressions and even walk.

Dates of note:

1921: Czech playwright Karel Čapek released a science fiction play "Rossum's Universal Robots" which introduced the idea of "artificial people" which he named robots. This was the first known use of the word.

1929: Japanese professor Makoto Nishimura built the first Japanese robot, named Gakutensoku.

1949: Computer scientist Edmund Callis Berkley published the book "Giant Brains, or Machines that Think" which compared the newer models of computers to human brains.

Birth of AI: 1950-1956

This range of time was when the interest in AI really came to a head. Alan Turing published his work "Computer Machinery and Intelligence" which eventually became The Turing Test, which experts used to measure computer intelligence. The term "artificial intelligence" was coined and came into popular use.

Dates of note:

1950: Alan Turing published "Computer Machinery and Intelligence" which proposed a test of machine intelligence called The Imitation Game.

1952: A computer scientist named Arthur Samuel developed a program to play checkers, which is the first to ever learn the game independently.

1955: John McCarthy held a workshop at Dartmouth on "artificial intelligence" which is the first use of the word, and how it came into popular usage.

AI maturation: 1957-1979

The time between when the phrase "artificial intelligence" was created, and the 1980s was a period of both rapid growth and struggle for AI research. The late 1950s through the 1960s was a time of creation. From programming languages

that are still in use to this day to books and films that explored the idea of robots, AI became a mainstream idea quickly.

The 1970s showed similar improvements, such as the first anthropomorphic robot being built in Japan, to the first example of an autonomous vehicle being built by an engineering grad student. However, it was also a time of struggle for AI research, as the U.S. government showed little interest in continuing to fund AI research.

Notable dates include:

1958: John McCarthy created LISP (acronym for List Processing), the first programming language for AI research, which is still in popular use to this day.

1959: Arthur Samuel created the term "machine learning" when doing a speech about teaching machines to play chess better than the humans who programmed them.

1961: The first industrial robot Unimate started working on an assembly line at General Motors in New Jersey, tasked with transporting die casings and welding parts on cars (which was deemed too dangerous for humans).

1965: Edward Feigenbaum and Joshua Lederberg created the first "expert system" which was a form of AI programmed to replicate the thinking and decision-making abilities of human experts.

1966: Joseph Weizenbaum created the first "chatterbot" (later shortened to chatbot), ELIZA, a mock psychotherapist, that used natural language processing (NLP) to converse with humans.1968: Soviet mathematician Alexey Ivakhnenko published "Group Method of Data Handling" in the journal "Avtomatika," which proposed a new approach to AI that would later become what we now know as "Deep Learning."

1973: An applied mathematician named James Lighthill gave a report to the British Science Council, underlining that strides were not as impressive as those that had been promised by scientists, which led to much-reduced support and funding for AI research from the British government.

1979: James L. Adams created The Standford Cart in 1961, which became one of the first examples of an autonomous vehicle. In '79, it successfully navigated a room full of chairs without human interference.

1979: The American Association of Artificial Intelligence which is now known as the Association for the Advancement of Artificial Intelligence (AAAI) was founded.

AI agents: 1993-2011

Despite the lack of funding during the AI Winter, the early 90s showed some impressive strides forward in AI research, including the introduction of the first AI system that could beat a reigning world champion chess player. This era also saw early examples of AI agents in research settings, as well as the introduction of AI into everyday life via innovations such as the first Roomba and the first commercially-available speech recognition software on Windows computers.

The surge in interest was followed by a surge in funding for research, which allowed even more progress to be made.

Notable dates include:

1997: Deep Blue (developed by IBM) beat the world chess champion, Gary Kasparov, in a highly-publicized match, becoming the first program to beat a human chess champion.

1997: Windows released a speech recognition software (developed by Dragon Systems).

2000: Professor Cynthia Breazeal developed the first robot that could simulate human emotions with its face,which included eyes, eyebrows, ears, and a mouth. It was called Kismet.

2002: The first Roomba was released.

2003: Nasa landed two rovers onto Mars (Spirit and Opportunity) and they navigated the surface of the planet without human intervention.

2006: Companies such as Twitter, Facebook, and Netflix started utilizing AI as a part of their advertising and user experience (UX) algorithms.

2010: Microsoft launched the Xbox 360 Kinect, the first gaming hardware designed to track body movement and translate it into gaming directions.

2011: An NLP computer programmed to answer questions named Watson (created by IBM) won Jeopardy against two former champions in a televised game.

2011: Apple released Siri, the first popular virtual assistant.

Top 20 Applications of Artificial Intelligence (AI) in 2025

1. E-Commerce

Artificial Intelligence in the field of E-commerce helps the organization to establish a good engagement between the user and the company. AI chatbots are used to provide customer support instantly and help to reduce complaints and queries to a great extent. Let's take a closer look at AI applications in E-commerce.

Personalization: Using this feature, customers would be able to see those products based on their interest pattern and that eventually will drive more conversions.

Dynamic Pricing Structure: It's a smart way of fluctuating the price of any given product by analyzing data from different sources and based on which price prediction is being done.

Voice Search: With the introduction of this feature, many applications and websites are using voice-over searches in their system.

2. Education Purpose

Educational sectors were totally organized and managed by human involvement till some years back. But these days, the educational sector is also coming under the influence of Artificial Intelligence.

Voice Assistant: With the help of AI algorithms, this feature can be used in multiple ways to save time and provide help to assist users when required.

Enhanced Engagement and Accessibility : It helps in the development of interactive learning environments, support engagement with features such as gamification and provide accessible resources for students with disabilities.

Smart Content Creation: **It uses algorithms to detect, predict and design content and provide** valuable insights based on the user's interest which can include videos, audio, infographics etc.

3. Robotics

Artificial Intelligence is one of the major technologies that provide the robotics field with a boost to increase their efficiency. AI provides robots to make decisions in real time and increase productivity.

Autonomous Navigation: AI equips robots including self-driving cars and drones to navigate through complex environments by using sensors and computer vision, making decisions based on their surroundings.

Healthcare Robotics: Robots powered by AI assist in surgeries and patient care bring precision to the medical procedures and support in recovery.

Industrial Automation: **Robots automate repetitive tasks, carry out quality control and optimize** assembly processes by learning from experience to improve their performance.

4. GPS and Navigations

GPS technology uses AI to provide the best available route to the users for traveling. It helps the user to choose their type of lane and roads which increases the safety features of a user.

Voice Assistance: This feature allows users to interact with the AI using a hands-free feature which allows them to drive while communicating with the navigation system.

Dynamic Route Optimization: AI reads real-time data to ensure optimized routes based on the current traffic conditions so that the navigation is faster and efficient.

Predictive Traffic Analysis: Using historical data and machine learning, predictions enable the navigation app to suggest for the best possible time to travel.

5. Healthcare

Various algorithms are used to build precise machines that are able to detect diseases inside the human body. Artificial Intelligence is also used to find the current vacant beds in the hospitals of a city that saves the time of patients who are in emergency conditions.

Medical Imaging and Diagnostics: AI algorithms detect problems like cancer, heart disease and neurological problems from medical images such as X-rays, MRIs and CT scans. AI helps doctors determine patterns that human doctors may miss, thus improving their accuracy in diagnosing patients.

Predictive Analytics: AI models predict the likelihood of future conditions such as diabetes, strokes, or heart attacks by analyzing patient data including medical histories, lifestyle and genetic information.

Virtual Health Assistants: AI-powered chatbots and virtual assistants are used for patient support, scheduling appointments and monitoring chronic conditions. They provide 24/7 assistance reducing the burden on healthcare professionals.

6. Automobiles

It makes the automotive industry vehicles smarter, safer and more efficient. AI technologies are transform how cars are designed and driven.

Predictive Maintenance: AI can predict when a vehicle's components require maintenance or replacement. It can forecast issues before they result in breakdowns by analyzing data from sensors embedded in the vehicle.

V2X Communication: V2X communication with the help of AI helps cars anticipate hazards, reduces accidents and optimizes the management of traffic-carrying functions especially in crowded urban areas.

Driver Behavior Monitoring: AI shall monitor a driver's behavior via sensors and cameras to identify signals of fatigue or distraction. Using that, the system shall provide a warning or adjust the driving conditions for improvement towards safety.

7. Agriculture

It is used to detect various parameters such as the amount of water and moisture, amount of deficient nutrients etc in the soil. There is also a machine that uses AI to detect where the weeds are growing and the soil is infertile.

Stock Monitoring: Rigorous monitoring to ensure that crops that not being affected by any disease, AI uses Computer Vision to check crop feeds live and alarm when any abnormality arises.

Pest Management: Algorithms can analyze data from multiple sources to identify early warnings to respective farmers. This technology also enables less usage of harmful pesticides by offering the best resources for pest management.

Forecasting: With the help of AI analyzing the weather forecast and crop growth has become more convenient in the field of agriculture and the algorithms help farmers to grow crops with effective business decisions.

8. Human Resource

The online selection processes are done using the voice and camera permission of the candidate's device. Here Artificial Intelligence is used to detect any kind of malpractice behavior and many other things.

Screening: The screening process of candidates can be automated to reduce the timeframe and eliminate excess resources for this task. This helps shortlist potential candidates and resumes by assessing insightful information in no time.

Onboarding: Offering a personalized onboarding kit has become famous in many tech companies and this enables HR professionals to complete the initial paperwork easily.

Workforce Planning: To predict future work plans and align resources, AI can be helpful in the human resource sector. This requires a predictive analysis based on which workforce planning is being performed.

9. Lifestyle

Some examples of AI impact on lifestyle are spam filters in the mail, fraud call detection, face unlock of mobile in our mobile and laptops etc.

Virtual assistants: Assistants such as Siri, Alexa and Google Assistant can work on managing different tasks using only voice commands and deliver information reminding the user to do things and even controlling some smart home appliances.

Personalized Recommendations: AI analyses all the user behaviors and preferences offering recommendations in online shopping, streaming services like Netflix and Spotify and even social media feeds making them more interactive with the user.

Smart Home Devices: Smart home technology includes security systems, so that the control and monitoring of home environments are automatically done according to user habits and preferences.

10. Social media

Social media platform such as Facebook, Instagram uses Artificial Intelligence to show relevant content to the user.

Fraud Detection: AI uses algorithms to spot and remove any fake accounts that are associated with any social media platform. Fraudsters generally use those accounts to perform unethical activities.

Sentiment Analysis: People are more connected and likely to spend more time on social media platforms. AI uses its algorithm to determine the pattern, providing a better experience while maintaining awareness.

11. Gaming

Artificial Intelligence is used to make a human-like simulation in gaming. Various modern games use real-world simulation for gaming.

Adaptive Difficulty: The AI can adjust the level of difficulty of the game in real-time depending on a player's skill level. If he is struggling, AI can ease him and if it finds him going through easy rides through the levels, it can challenge him further for a balanced experience.

Personalized Experiences: AI algorithms analyze a player's behavior and preferences to tailor the game's experience. This could include personalized quests, recommendations for in-game items ensuring each player has a unique experience.

Fair Play: Online multiplayer games make use of AI to detect cheating by identifying patterns that might indicate behavior such as unnatural movement or impossible actions. It helps in the maintenance of fair play and integrity in competitive gaming.

12. Astronomy

AI is used to investigate galaxy mergers and stars to predict the future of human beings. The movement of stars in the sky is recorded for the whole day and night over a large time to read the changes.

Analysis: It uses various classifiers and identifier to extract insights by using time series.

Detection and Classification: AI uses its algorithms to predict incidents by performing analysis from exoplanets and other subsidies. They are well capable of performing various other astronomical patterns based on large datasets.

13. Chatbots

Chatbots are defined as a tool that is used to respond to the text that is given to them as input. The customer or user sends the query according to their need and the chatbot gives the most appropriate output to provide solution.

NLP: Natural Language Processing allows chatbots to interact with humans over chat in a more interactive way by offering quick resolution to their queries in no time. It also allows chatbots to process text or speech inputs and extract meaning to generate the desired response.

Multi-Language: AI-powered chatbots are capable of handling multiple languages to provide support to users worldwide.

Adaptation: Through machine learning techniques, chatbots can analyze and identify areas for improvement and update themselves.

14. Surveillance

It is also used in the field of surveillance by recognizing far faces and objects. This helps the military to protect their areas and prevent any attack in real time.

Object Detection: With the help of CV, AI algorithms track objects of interest in real time. This feature allows the system to detect any object's movement and further analysis.

Predictive Analysis: It can analyze historical data and patterns to make effective decisions. This can be achieved by empowering different techniques used in machine learning, deep learning and artificial intelligence.

Behavior Analysis: Surveillance can analyze human behavior patterns such as gesture recognition or body language analysis to assess potential threats or suspicious activities.

15. Finance

It is used to send automated tests and mail, predict risk in loans and detect any unwanted transactional activities of the targeted customers. Spam and fraud filters, uncharacterized actions, responding to threats, etc are the tools that help in the field.

Fraud Detection: By continuously learning from historical data, AI gets capable of handling and working with such datasets. It also enables the ability to prevent from any future threats.

Risk Assessment: AI algorithms help in safeguarding risk assessment models by using large datasets and identifying potential risks in real-time.

Forecasting: Financial planning and forecasting that can be done to create easy opportunities for businesses. This can be achieved by analyzing historical data, market trends and economic indicators.

16. Data Security

Data security is one of the major concerns for any tech company. There are many AI application that are used to keep these data safe and prevent from threats and attacks.

Threat Detection: AI algorithms can analyze and work on large datasets from various sources such as network logs, user behavior, system activities etc. This helps in the early detection of malware or any unknown threats.

Vulnerability Management: With the help of these algorithms, anyone having access can start the scanning process automatically for known vulnerabilities and prioritize them based on the potential impact.

Malware Detection: AI algorithms can predict patterns of malware to provide a shield against any new threats. By allowing machine learning and deep learning techniques, AI can easily identify previously unknown malware to prevent future attacks.

17. Travel and Transport

Travel routing and ride-sharing which used real-time traffic for a specific ride are possible due to the use of Artificial Intelligence.

Planning and Personalization: It uses customer's historic data along with the real-time data to offer any customized flexible plan under their budget. This helps in driving more sales and matches individual needs.

Predictive Analysis (Pricing): AI algorithms can analyze historical travel data, market trends and other variables. Companies use this strategy to offer accommodations and travel assistance at higher rates.

18. Marketing

AI is used to make the reach by targeting the appropriate audience for a particular product. Apart from that the chatbots are used to resolve the consumer issue in less time.

Segmentation: Algorithms to work on large sets of data to identify and visualize customer's pattern and their behavior. This also helps companies to segregate the user base on different metrics such as age, gender, purchasing history etc.

Content Creation: It uses NLP to generate content. It is also well capable of generating different segments of content in very less time.

Advertising: AI enhances advertising campaigns by optimizing targeting, bidding and creative elements that can boost their advertising strategy. It can also help in analyzing users sentiments based on different factors and datasets.

19. Entertainment

This field uses artificial intelligence for creating various contents. The entertainment industry also uses artificial intelligence for the categorization and classification of content according to the user. Let's take a closer look at AI applications in Entertainment.

Recommendation: It can analyze vast amounts of data that includes viewing patterns user preference and historic data. Then this data is used to provide personalized recommendations to keep up the engagement ratio and watch time.

Audience Insight: AI helps entertainment companies analyze large volumes of data and then decide mainstream content and make effective business decisions.

20. Military

The government is planning to use artificial intelligence for various military operational support. It will also help in automatic artilleries and weapons. Let's take a closer look at AI applications in the Military.

Decision Support: Algorithms can analyze large amounts of data including sensor inputs and historical information. These insights provide aid while taking a decision that includes stock management, resource allocation and so on without actual human intervention.

Cyberattack: AI plays a crucial role in detecting and responding to cyber threats in military networks. AI algorithms are capable enough to handle and manage vast datasets that can detect any abnormal activity before it actually occurs.

Training: Used in training military staff in different situations that can happen. This helps trainees in making the right decision at the moment and strategize their plan effectively.

In conclusion, the applications of AI are vast, impacting industries and daily life in many ways. AI is reshaping how we live and work. As technology continues to advance, the potential for AI to drive innovation and solve complex problems is immense, leading a future where intelligent systems enrich our lives.

AI agent and Types of AI Agents:

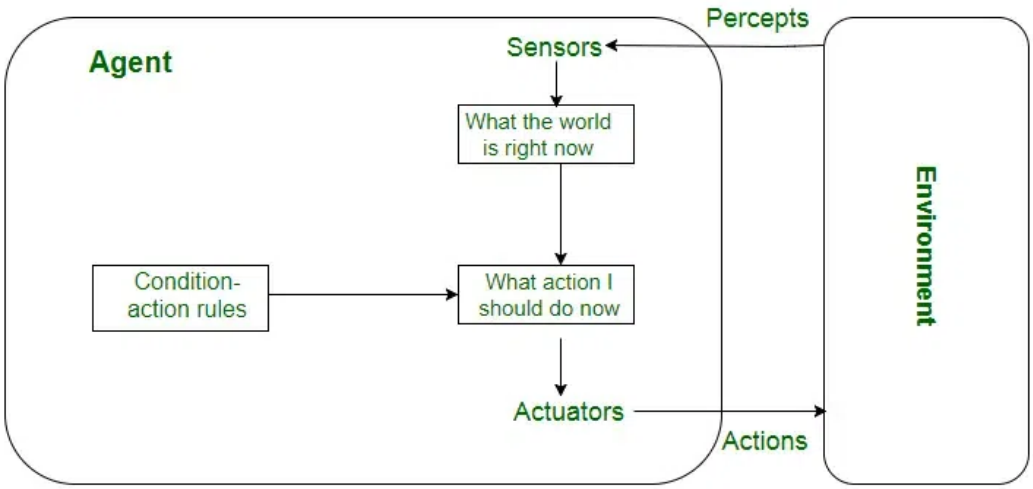
An AI agent is a system that perceives its environment, makes decisions, and takes actions to achieve goals, with types ranging from simple reflex agents to complex, learning agents that adapt over time. Common types include **simple reflex agents** (acting on current perceptions), **model-based agents** (using internal world models), **goal-based agents** (aiming for specific objectives), **utility-based agents** (optimizing actions for maximum reward), and **learning agents** (improving from experience). Other categorizations include multi-agent systems (interacting with other agents) and hierarchical agents (structured decision-making).

Types of AI Agents

AI agents are often categorized by their intelligence and capabilities:

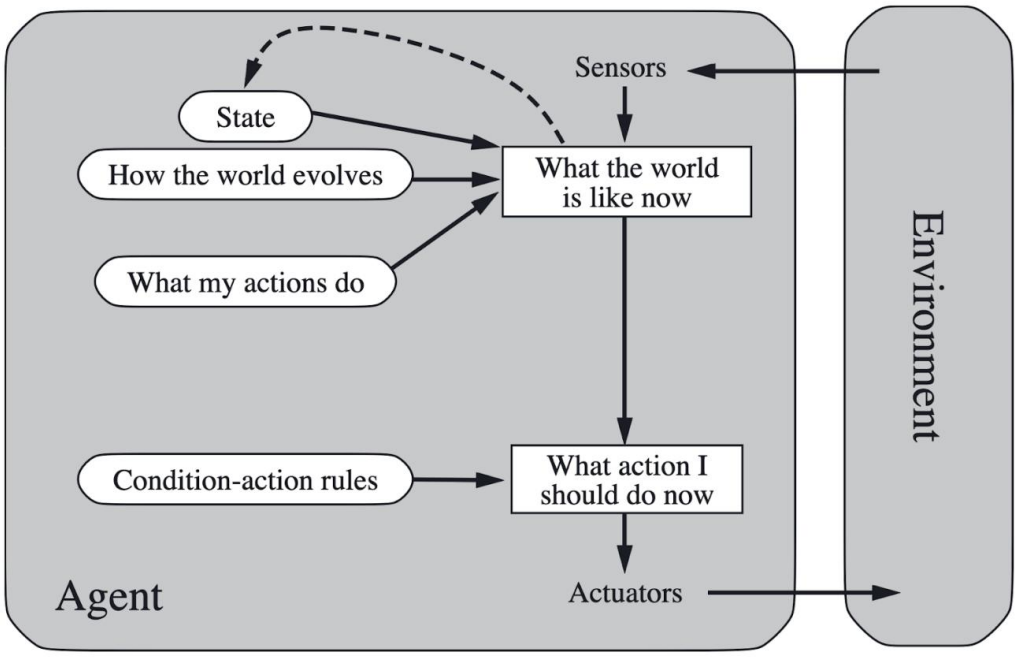
1. **Simple Reflex Agents**:

- **How they work**: They act based on their current perceptions using pre-defined "condition-action" rules without considering past experiences or having an internal model of the world.
- **Example**: A basic customer service chatbot that responds to simple questions with predefined answers.



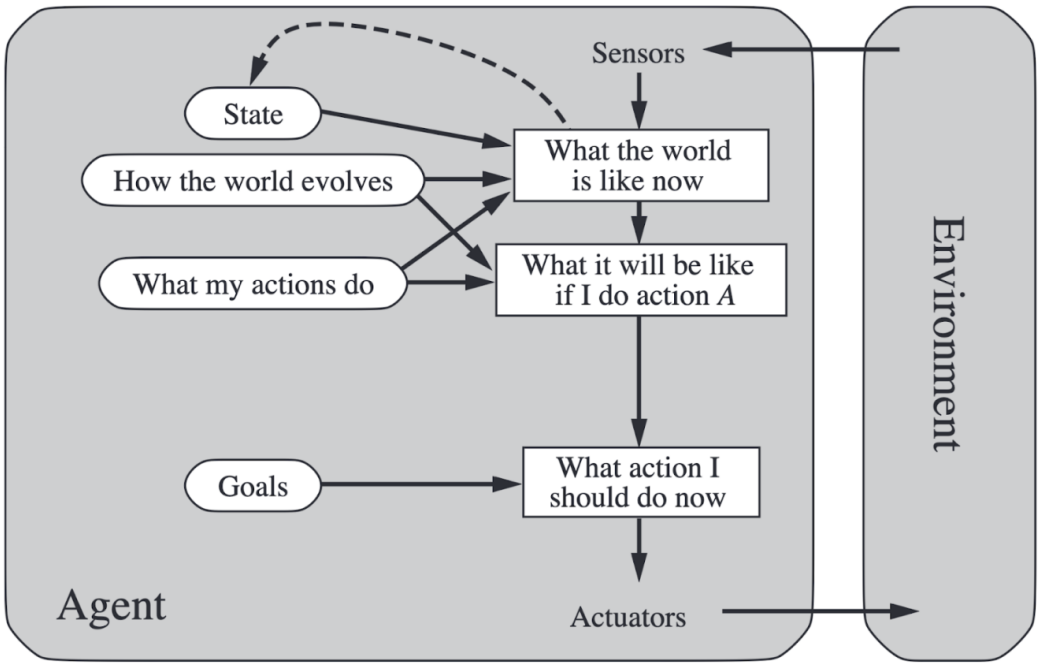
2. Model-Based Reflex Agents:

- **How they work**: These agents maintain an internal model or "world model" of the environment to keep track of what's happening, allowing them to understand context and "fill in the gaps" of missing information.
- **Example**: A self-driving car that uses sensors to build a model of its surroundings to make decisions.



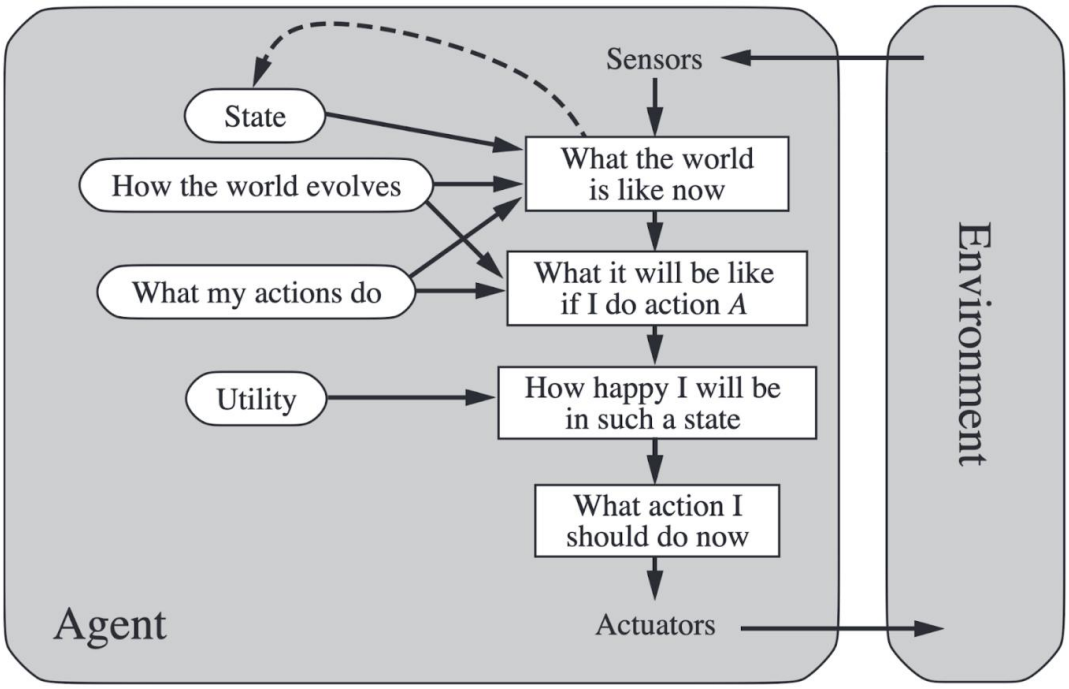
2. **Goal-Based Agents**:

- **How they work**: They have specific goals and use their internal world model to plan sequences of actions that will help them achieve those goals.
- **Example**: A robot that plans a path to a specific location in a warehouse.



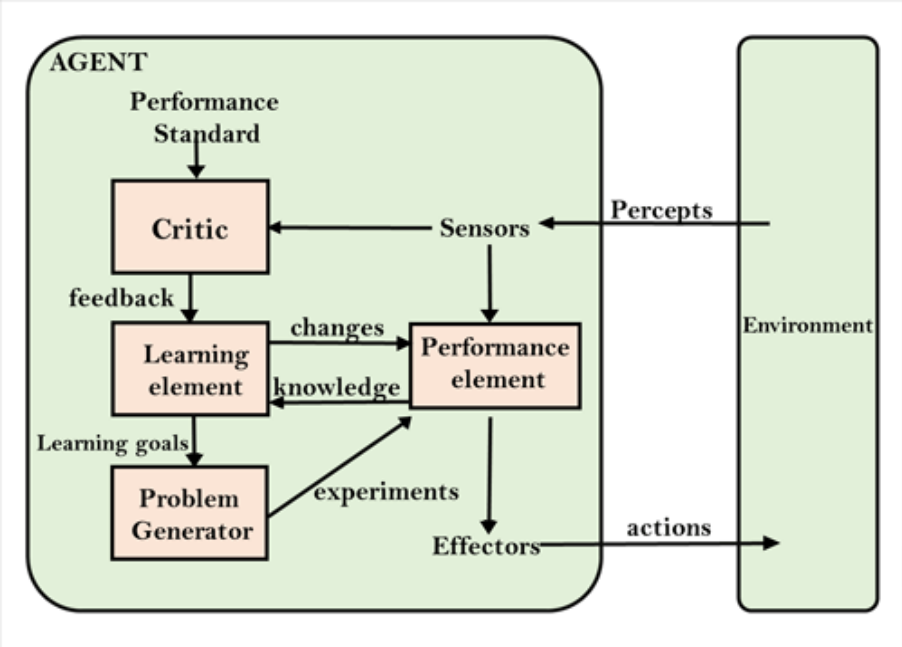
3. **Utility-Based Agents**:

- **How they work**: Beyond just achieving a goal, these agents aim to maximize a "utility" or reward function, evaluating different actions to find the one that provides the highest expected utility or satisfaction.
- **Example**: A financial trading bot that seeks to maximize profits by choosing actions that yield the highest returns.



4. **Learning Agents**:

- **How they work**: The most advanced type, learning agents can improve their performance over time by learning from their own experiences and feedback. They have a learning module, a performance system, and a critic to guide their learning.
- **Example**: A recommendation system that learns your preferences to provide better suggestions.



Problem Solving in Artificial Intelligence

Problem solving is a fundamental concept in artificial intelligence (AI) where systems are designed to identify challenges, make decisions and find efficient solutions. AI uses agents which are systems that perceive their environment and take actions to achieve specific goals. They go beyond simple reflex agents which react to stimuli based on pre-defined rules. Instead, problem-solving agents actively analyze situations, evaluate different options and choose the best action to reach their goal.

These agents work by:

Perceiving the environment: They collect data about their surroundings such as sensor inputs or observations.

Defining the problem: They clearly understand the problem including the starting point, the available actions and the desired goal.

Exploring different possibilities: They consider various ways to solve the problem and evaluate which approach is likely to succeed.

Evaluating and deciding: Once they explore options, they assess the outcomes and pick the best course of action based on factors like time, resources and success likelihood.

Learning and adapting: Many problem-solving agents can learn from past experiences, improving their decision-making abilities over time.

Types of Problems in AI

AI systems can face different types of problems. These problems are categorized based on their impact on the system's performance and how they can be handled:

1. Ignorable Problems

These are minor issues that have little or no effect on the AI system's overall performance. They are often harmless and do not require immediate attention.

Examples: Small inaccuracies in predictions that don't affect the final result like a slight error in image classification.

Minor issues in data pre-processing that don't change the outcome.

Handling: Ignored as they don't interfere with the system's primary function.

2. Recoverable Problems

Recoverable problems are those where the AI system encounters an issue but can be fixed with intervention either automatically or manually, such as error-handling functions.

Examples: Missing data that can be filled in using statistical methods.

System crashes that can be fixed by restoring from a backup.

Handling: Requires action like automated error handling, retraining the model or system recovery through checkpoints.

3. Irrecoverable Problems

These are severe issues that cause permanent damage or failure, making it impossible for the system to recover. They can lead to significant performance loss.

Examples: Corrupted training data that causes bias and reduces the model's effectiveness.

Adversarial attacks that make the model untrustworthy.

Overfitting, where the model becomes too specialized and cannot adapt to new data.

Handling: Requires a complete rebuild of the system including retraining the model or updating the data to eliminate underlying issues.

Steps in Problem Solving in AI

AI problem solving follows a structured, logical process similar to human thinking, it's steps include:

Problem Definition: First, the problem needs to be clearly defined. This includes understanding the inputs, constraints and what the solution should look like.

Problem Analysis: Once the problem is defined, it's analyzed in more detail. This helps in understanding its limitations and possible solutions.

Knowledge Representation: All the important information is organized so the AI can understand and work with it. This could include creating graphs or using databases.

Problem Solving: AI uses appropriate methods to solve the problem. This often means comparing different strategies to find the most efficient one.

Testing and Evaluation: Finally, after the solution is implemented, testing and evaluation ensure the solution meets all requirements and performs as expected.

Key Components of Problem Formulation in AI

Effective problem-solving in AI is dependent on several important components:

Initial State: This is the starting point of the problem where the AI begins its process. It sets the context and helps identify how the agent will approach the challenge.

Action: At this stage, AI identifies all the possible actions it can take from the initial state. Each action has an impact on how the system moves closer to solving the problem.

Transition: This refers to how the system changes from one state to another after an action is taken. Transition modeling helps show how the agent's actions influence the next steps.

Goal Test: Once an action is taken, AI checks if it has reached its goal. If the goal is achieved, the problem-solving process stops and the solution is considered complete.

Cost Function: This step assigns a numerical value to the cost of achieving the goal. The cost can include resources like time, energy or money and helps decide the most efficient way to reach the goal.

Artificial Intelligence	Machine Learning	Deep Learning
AI stands for Artificial Intelligence, and is basically the study/process which enables machines to mimic human behaviour through particular algorithm.	ML stands for Machine Learning, and is the study that uses statistical methods enabling machines to improve with experience.	DL stands for Deep Learning, and is the study that makes use of Neural Networks(similar to neurons present in human brain) to imitate functionality just like a human brain.
AI is the broader family consisting of ML and DL as it's components.	ML is the subset of AI.	DL is the subset of ML.
AI is a computer algorithm which exhibits intelligence through decision	ML is an AI algorithm which allows system to learn from data.	DL is a ML algorithm that uses deep(more than one layer) neural networks to analyze data and provide

Artificial Intelligence	Machine Learning	Deep Learning
making.		output accordingly.
Search Trees and much complex math is involved in AI.	If you have a clear idea about the logic(math) involved in behind and you can visualize the complex functionalities like K-Mean, Support Vector Machines, etc., then it defines the ML aspect.	If you are clear about the math involved in it but don't have idea about the features, so you break the complex functionalities into linear/lower dimension features by adding more layers, then it defines the DL aspect.
The aim is to basically increase chances of success and not accuracy.	The aim is to increase accuracy not caring much about the success ratio.	It attains the highest rank in terms of accuracy when it is trained with large amount of data.
Three broad categories/types Of AI are: Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI)	Three broad categories/types Of ML are: Supervised Learning, Unsupervised Learning and Reinforcement Learning	DL can be considered as neural networks with a large number of parameters layers lying in one of the four fundamental network architectures: Unsupervised Pre-trained Networks, Convolutional Neural Networks, Recurrent Neural Networks and Recursive Neural Networks
The efficiency Of AI is basically the efficiency provided by ML and DL respectively.	Less efficient than DL as it can't work for longer dimensions or higher amount of data.	More powerful than ML as it can easily work for larger sets of data.
Examples of AI applications include: Google's AI-Powered Predictions, Ridesharing Apps Like Uber and Lyft, Commercial Flights Use an AI	Examples of ML applications include: Virtual Personal Assistants: Siri, Alexa, Google, etc., Email Spam and Malware Filtering.	Examples of DL applications include: Sentiment based news aggregation, Image analysis and caption generation, etc.

Artificial Intelligence	Machine Learning	Deep Learning
Autopilot, etc.		
AI refers to the broad field of computer science that focuses on creating intelligent machines that can perform tasks that would normally require human intelligence, such as reasoning, perception, and decision-making.	ML is a subset of AI that focuses on developing algorithms that can learn from data and improve their performance over time without being explicitly programmed.	DL is a subset of ML that focuses on developing deep neural networks that can automatically learn and extract features from data.
AI can be further broken down into various subfields such as robotics, natural language processing, computer vision, expert systems, and more.	ML algorithms can be categorized as supervised, unsupervised, or reinforcement learning. In supervised learning, the algorithm is trained on labeled data, where the desired output is known. In unsupervised learning, the algorithm is trained on unlabeled data, where the desired output is unknown.	DL algorithms are inspired by the structure and function of the human brain, and they are particularly well-suited to tasks such as image and speech recognition.
AI systems can be rule-based, knowledge-based, or data-driven.	In reinforcement learning, the algorithm learns by trial and error, receiving feedback in the form of rewards or punishments.	DL networks consist of multiple layers of interconnected neurons that process data in a hierarchical manner, allowing them to learn increasingly complex representations of the data.

What is the Turing Test?

The Turing Test is a widely recognized benchmark for evaluating a machine's ability to demonstrate human-like intelligence. The core idea is simple: A

human judge engages in a text-based conversation with both a human and a machine. The judge's task is to determine which participant is human and which is the machine. If the judge is unable to distinguish between the human and the machine based solely on the conversation, the machine is said to have passed the Turing Test.

Criteria for the Turing Test

The Turing Test does not require the machine to be correct or logical in its responses but rather to be convincing in simulating human conversation. The test is fundamentally about deception—the machine must fool the judge into believing that it is human.

The key criteria include the following for turing test:

Natural Language Processing (NLP): The machine must understand and generate human language fluently.

Knowledge Representation: The machine needs to handle and manipulate knowledge to provide contextually relevant responses.

Reasoning: The machine should demonstrate some form of logical reasoning, even if flawed, to sustain a conversation.

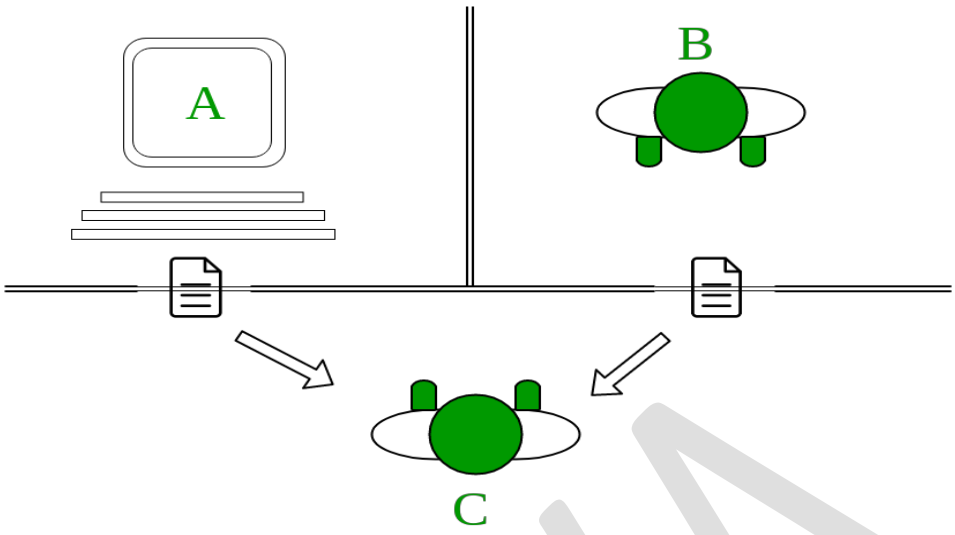
Learning: Ideally, the machine should learn from the interaction, adapting its responses over time.

How the Turing Test Works?

In a typical Turing Test scenario, three participants are involved: two humans and one machine.

The interrogator, a human judge, is isolated from the other two participants. The judge asks questions to both the human and the machine, aiming to identify which one is the human. The machine's goal is to respond in a way that makes it indistinguishable from the human participant. If the judge cannot reliably identify the machine, the machine is considered to have passed the Turing Test.

SHIVA



SHIVA

The first step is to define the Sudoku puzzle as a 9x9 grid where 0 represents an empty cell. We also define a function print_sudoku to display the puzzle in a human readable format.

```
Initial Sudoku Puzzle:

5 3 0  | 0 7 0  | 0 0 0
6 0 0  | 1 9 5  | 0 0 0
0 9 8  | 0 0 0  | 0 6 0
- - - - - - - - - - - -
8 0 0  | 0 6 0  | 0 0 3
4 0 0  | 8 0 3  | 0 0 1
7 0 0  | 0 2 0  | 0 0 6
- - - - - - - - - - - -
0 6 0  | 0 0 0  | 2 8 0
0 0 0  | 4 1 9  | 0 0 5
0 0 0  | 0 8 0  | 0 7 9
```

SHIVA